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Initial Post

by [Ali Alzahmi](#) - Tuesday, 26 August 2025, 7:12 PM

The spread of multi-agent systems (MAS) in artificial intelligence has reflected the usefulness of organised communications. Not only the agents need to be able to share data but also intentions, goals and commitments. In order to meet this requirement, ACLs, in particular, the Knowledge Query and Manipulation Language (KQML) were invented, the protocol of which is based on the speech act theory (Kim et al., 2024). Unlike the case with regular programming calls, ACLs are concerned with the semantics of interaction and not with its syntax. The optimal merit of ACLs is that it confines a heterogeneous environment (Wang et al., 2025). The agents that have been programmed in different programming languages and which are designed to run on different systems will be capable of communicating with each other, provided they work with the same type of communication standard. In addition, ACLs can include semantically rich messages, as messages are performatives (except inform, request), e.g. converse or play-game, and the intention of the agent is made evident. Finally, systems that incorporate the use of ACLs scale better, since effectively the communication patterns do not have to undergo vast reengineering as more agents are added to the system. Along with their benefits, ACLs are disadvantaged in practice as well. They create the necessity of doing significant reasoning to the agents, rising the complexity of computer calculations. There can also be a given efficiency restriction: ACLs are also overhead relative to lightweight calls and can thus not be applied in real-time or performance-based applications (Younas et al., 2025). These drawbacks explain why ACLs remain most common in academic use cases, when compared to commercial use cases, where less complex mechanisms are more popular, such as an API or message queues. In Python or Java, calling methods are a simple, high-performance and predictable way of communicating between software components (Dong et al., 2025). They best suited suited in tightly coupled systems. The preference of ACLs, in their turn, is in dynamic open environments, where agents are subject to autonomous action that necessitates the interpretation of the goals of other agents. This is due to the fact that, though ACLs, such as KQML are not heavily used in the industry, they are also a major research contribution.

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Peer Response

by [Abdulla Alshaibani](#) - Wednesday, 3 September 2025, 6:59 PM

Hi Ali,

Thank you for sharing such a detailed perspective on ACLs and their role in multi-agent systems. I really appreciate how you highlighted the distinction between semantic-rich communication and lightweight method calls in languages like Python or Java. Your observation that ACLs are more common in research while APIs and message queues dominate in commercial environments is very accurate and worth deeper reflection.

One area where ACLs have moved closer to practical use is in energy and utility management. For example, Siemens has experimented with agent-based smart grid systems where ACL-inspired communication enables negotiation between power producers and consumers(Roche et al., 2010). In such dynamic environments, lightweight APIs may struggle to capture intent, while ACLs provide richer semantics for resource balancing. Similarly, NASA's autonomous spacecraft systems use agent communication models rooted in ACL concepts to delegate tasks and interpret goals among distributed components something not achievable with simple function calls(Pell et al., 1997).

Your point on computational overhead is valid, and it explains why industries often prefer APIs for speed and reliability. However, as open, dynamic, and distributed systems become more common like autonomous vehicles and supply chain networks ACL-style frameworks may gain broader adoption despite the overhead(Garikapati and Shetiya, 2024).

Q: Do you think organizations will adapt ACLs for practical commercial use as these dynamic systems expand, or will they continue to rely on simplified alternatives like APIs?

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Peer Response

by [Ana-Maria Stoica](#) - Friday, 5 September 2025, 4:23 PM

Hi Ali,

Your text is a strong one in terms of accuracy, there is a clear understanding of some core ideas, for example the differences between regular programming methods and ACLs. I particularly appreciate that you highlighted the idea of semantics versus syntax. The structure is clear and concise, with well-defined delimitations between your ideas. However, in some cases there is a lack of clarity. For example, I had a hard time grasping your phrase "confines a heterogeneous environment," and some other syntax choices affect the text's readability.

I agree with your point regarding the limited industrial use of ACLs. KQML and FIPA-ACL were developed to enable complex communications and make agents capable of requesting or negotiating. FIPA-ACL slowly replaced KQML because the latter lacked semantics (Drasko and Rakic, 2024). Such is the case with non-ACL protocols, which are widely used because they are effective without extra work and high computational costs. It all comes down to the goal, and in industrial settings, simpler protocols like MQTT, AMQP, or CoAP are often preferred because they focus on delivering data and efficiency rather than understanding the meaning.

I would like to add to this that still, there is interest in adapting ACLs in an industry context. For example, Jin and Chen (2020) developed a multi-agent rotating machinery diagnostic system using a KQML-based communication strategy. The results show a clear advantage of using this approach, the accuracy and stability of rotating machinery diagnosis were both improved. This is a clear illustration of how using ACLs might benefit the more complex industrial field, but, as the same article points out, external factors and the many difficulties of actual operations are not fully taken into account in experimental settings, which I believe is a strong argument for your initial point.

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Peer Response

by [Fahad Abdallah](#) - Saturday, 6 September 2025, 7:28 AM

Hi Ali, your post about Agent Communication Languages (ACLs) in multi-agent systems was interesting and valuable. You have also pointed out clearly the importance of ACLs in allowing agents to share not only information but also intentions, goals, and commitments, which is essential in systems acting in diverse and dynamic environments. Your focus on the semantic quality of ACLs was especially appealing to me because it indicates their exceptional worth compared to simple syntactic method calls. Such an attitude is consistent with contemporary surveys of multi-agent systems, which note that heterogeneous agent communication is a key to coordination and collective intelligence (Guo et al., 2024).

I also liked your scalability argument. As you said, new agents can be easily integrated via ACLs without making significant changes in how new communication protocols are designed, which is essential to an open and adaptable environment. This is aligned with the recent literature on multi-agent reinforcement learning, in which interaction between agents substantially increases cooperation and system performance (Zhu et al., 2024). Performatives like inform, request, or play-game help show how ACLs can learn the logic and reasoning framework of actions taken by agents, which is key to more intelligent distributed systems (Calegari et al., 2021).

Meanwhile, I would have to say I like the fact that you recognised the limitations. Computational and reasoning complexity of ACLs is still a barrier, particularly in real-time or resource-constrained applications. This is one of the reasons why industries tend to use less complex and efficient communication systems such as APIs or message queues. However, I believe that an exciting research direction to explore in the future is the fusion of ACL concepts with contemporary engines, including large language model-driven multi-agent systems. The methods would allow finding a balance between semantic richness and practical efficiency, and the ACL-inspired communication would become more feasible in the application.

Your discussion has generally given a balanced perspective of ACLs' strengths and weaknesses, and your insights align with the emerging research directions.

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Peer Response

by [Ali Alhammadi](#) - Monday, 8 September 2025, 7:13 PM

Ali, your post shows clearly how Agent Communication Languages (ACLs) like KQML deal with the fact that agents need to communicate not only data but also intentions, goals, and commitments. I concur with you that emphasizing semantics, compared to syntax, is a key to helping multi-agent systems (MAS) reach more meaningful levels of interaction. It also finds manifestation in the recent work on large language models, in which it is pointed out that they can effectively serve as hierarchical coordination mechanisms capable of modeling more intricate communicative behaviors and assist in more elaborate cooperation between agents (Liu et al., 2023).

Your account of ACLs' capacity to deal with heterogeneous environments fascinated me. As you mentioned, multi-agent systems are usually characterized by components written in various programming languages or executed on different platforms. To make these agents cooperate, ACLs create a single communication standard. This reflects more general debates in foundation model literature, in which interoperability and scalability are crucial to creating flexible systems capable of accommodating the heterogeneous range of tasks and domains (Bommasani, 2021).

The other point you raised is quite critical and related to semantic richness. ACLs enhance the representation of agents' internal states by encoding performatives, including requests or commitments. Current theory of mind research on multi-agent collaboration notes the importance of semantically rich communication in the development of cooperative intelligence by demonstrating that large language models can infer and predict the intentions of other agents (Li et al., 2023).

Last but not least, I concur with your trade-off assessment. Whereas method invocation in a language such as Python or Java offers simplicity and the efficiency of tightly coupled systems, ACLs are better in open and dynamic environments where intent is as valuable as data transfer. In general, your analysis does an excellent job of providing a balance between the benefits and constraints of ACLs, including both practical issues and the long-term implications of this research.

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